W-F-406D
August 10, 1987
SUPERSEDING
W-F-406C
February 1, 1984

FEDERAL SPECIFICATION

FITTINGS FOR CABLE, POWER, ELECTRICAL AND CONDUIT, METAL, FLEXIBLE

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers box connectors for flexible conduit, armored cable, and nonmetallic sheathed cable, bushings for armored cable, couplings for flexible conduit, adapters for connecting flexible conduit to rigid conduit, and electrical metallic tubing, and clamps for grounding armored cable and ground wire

1.2 Classification.

1.2.1 Types, classes, styles, and kinds. Fittings furnished under this specification shall be of the following types, classes, styles, and kinds as specified (see 6.2 and 6.3):

Type I - Box connector, electrical

Class 1 - Flexible steel conduit and round armored cable
Style A - Straight connector, single attachment
Style B - Straight connector, duplex attachment
Style C - 45 degrees (°) angle connector
Style D - 90° angle connector

Kind a - Screw-in internal
Kind b - Screw-in external
Kind c - Clamp 2-screw
Kind d - Saddle clamp 1-screw
Kind e - Set screw
Kind f - Squeeze

AMSC N/A FSC 5975

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:26Z
Check the source to verify that this is the current version before use.

Class 2 - Oval (Flat) Armored Cable
Style E - Straight connector, single attachment
Style F - Straight connector, duplex attachment
Style G - 45° angle connector
Style H - 90° angle connector

Kind g - Screw-in external
Kind h - Clamp 2 screw
Kind i - Saddle clamp 1-screw
Kind j - Set-screw
Kind k - squeeze

Class 3 - Liquid-tight flexible metallic conduit connectors
Style I - Straight connector
Style J - 45° connector
Style K - 90° angle connector
Style L - Straight connector, male bashing style

Kind l - Screw-in external

Class 4 - Nonmetallic sheathed cable and service entrance cable connectors
Style M - Straight connector, single cable
Style N - Straight connector, duplex cable
Style O - 90° angle connector

Kind m - Oval cable
Kind n - Round cable

Type II - Bushing, electrical conductor, antishort, armored cable
Type III - Coupling, electrical conduit, flexible steel
Type IV - Adapter, electrical conduit

Class 5 - Flexible steel conduit or armored cable to rigid condit
Style P - With internal threaded connection to rigid conduit
Style Q - With threadless connection to rigid conduit

Class 6 - Flexible steel conduit or armored cable to electrical metallic tubing

Type V - Clamp, electrical grounding

Class 7 - With clamping jaws to fit standard iron pipe size (IPS) pipe
Style R - For bare armored cable
Style S - For bare or insulated wires

Class 8 - Armored cable hub for bare armored cable, with one bolthole tongue

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:26Z
Check the source to verify that this is the current version before use.

1.2.2 Sizes. Fittings furnished under this specification shall conform to standard commercial sizes, as specified (see 6.2 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

FED-STD-123 - Marking for Shipment (Civilian Agencies)
FED-STD-H28 - Screw threads for Federal Services

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:26Z
Check the source to verify that this is the current version before use.

2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitations.

National Electrical Manufacturers Association (NEMA):

FB1-1983 - Fittings, cast metal boxes and conduit bodies for conduit and cable assemblies

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

Underwriters Laboratories Inc. (UL):

UL - 467 - Standard for Grounding and Bonding Equipment
UL - 514B - Fittings for Conduit and Outlet Boxes

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Fittings covered by this specification shall be made of metal and conform to the applicable requirements of NEMA FB1, UL 514B and UL 467.

3.2 First article. When specified (see 6.2), the contractor shall furnish one fitting of the type, class, kind, and style as specified for the first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard compliance. Fittings shall conform to the applicable requirements of NEMA FB1, UL 467, or UL 514B (see 4.1.2).

3.3.1 Certification. Prior to approval of the first article, if one is submitted, or prior to approval of the first shipment, the contractor shall make available satisfactory evidence to the contracting officer or his authorized representative that the fittings be proposes to supply under this specification meets the requirements of NEMA FB1, UL 467, or UL 514B.

Source: https://assist.dla.mil --Downloaded: 2026-10-26T11:26Z
Check the source to verify that this is the current version before use.

3.3.2 UL certification. Acceptable evidence of meeting the requirements of UL 467 or UL 514B shall be the UL label, listing in the UL electrical construction materials directory, or a certified test report from a nationally recognized independent testing laboratory, acceptable to the contracting officer, stating that the fittings offered have been tested and conforms to UL 467 or UL 514B.

3.4 Components.

3.4.1 Box connectors. Box connectors shall be provided with a clamp for connecting to the conduit or cable and a hub and locknut arrangement for connecting to outlet boxes through standard conduit knockouts. Connectors that have tapered threads for threaded hubs need not have locknuts. Clamps shall neither pierce the armor, damage the insulation of cables, nor distort or damage the conduit when accomplishing its normal function. Attachment means shall provide an effective electrical bond between joined metallic parts. Hubs on connectors of trade size 1/2 inch or smaller shall fit the standard 1/2 inch conduit knockout. All other connector hubs shall fit conduit knockouts of the same size as the trade size of the connector. Screws shall be of the fillister head type except where required by fitting design, roundhead or washer head type screws may be used. The clamps or clamping means for attaching flexible conduit, armored cable, and nonmetallic sheathed cable to the connector shall provide adequate opening and clamping range to fulfill holding and bending requirements as specified in UL 514B and NEMA FB1.

3.4.2 Sleeves and bushings. Antishort sleeves and insulating bushings shall be unaffected by ordinary atmospheric conditions and be so formed that they shall not be distorted or dislodged by ordinary handling when secured in place.

3.4.3 Couplings and adapters. Couplings and adapters shall have a positive attachment means at each end and shall provide an effective electrical bond between joined metallic parts. Coupling and adapter sizes indicated by a single dimension are for use with the same size conduit on each end.

3.4.4 Grounding clamps. Ground clamps shall meet construction and test requirements of UL 467.

3.4.5 Finish. Fittings shall have surfaces which are not chipped, warped, cracked, or blistered. Surfaces which are normally in contact with conductor insulation shall be rounded to prevent abrasion or cutting of insulating material.

3.4.6 Threads. External and internal threads of IPS shall be of the straight or taper type, suitable for fitting use in accordance with FED-STD-H28.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:26Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on fittings when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4, and the tests of 4.5, and when specified, the first article pack inspection of 4.6 (see 4.6 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. All fittings offered for delivery at one time shall be considered a lot for the purpose of inspection. The inspection level shall be level II and the Acceptable Quality Level (AQL) shall be 4.0 percent

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:26Z
Check the source to verify that this is the current version before use.

defective. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.4 Examination. Each fitting shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests. Testing shall be as specified in NEMA FB1, UL 467, or UL 514B.

4.6 Preparation for delivery inspection. The inspection of the preservation, packaging, packing, and marking shall be in accordance with the requirements of section 4 of MIL-E-17555. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packing. Preservation and packing shall be in accordance with the requirements of MIL-E-17555 with the level of preservation and the level of packing as specified (see 6.2). Unit pack quantities shall be as specified (see 6.2).

5.2 Marking.

5.2.1 Military agencies. Shipments to military agencies shall be marked in accordance with MIL-STD-129.

5.2.2 Civil agencies. Shipments to civil agencies shall be marked in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The fittings covered by this specification are intended for use as follows:

Type I - For attaching cable or flexible conduit to an outlet box
Type II - To provide protection for the cable insulation at the point on an armored cable where the armor terminates

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:26Z
Check the source to verify that this is the current version before use.

Type III - To couple two sections of the same size flexible conduit
Type IV - For connecting armored cable or flexible conduit to rigid conduit or electrical metallic tubing
Type V - To provide means for connecting a ground wire, bare armored or insulated, to a grounded piping system

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type, class, style, kind, and trade size required (see 1.2.1 and 1.2.2).
c. When a first article inspection is required (see 3.2, 4.2.1 and 6.4).
d. When a first article pack inspection is required and time frame required for submission (4.2.1 and 4.6).
e. Level of preservation, packing required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements (see 1.2.1 and 1.2.2).

6.4 First article. When a first article inspection is required, (see 6.2) the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one fitting of each type. The contracting officer should include specific instructions in procurement documents regarding arrangements for examination, test, and approval of the first article (see 3.2).

6.5 Subject term (key word) listing.

Box connector
Cable
Conduit

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Source: https://assist.dla.mil --Downloaded: 2026-10-26T11:26Z
Check the source to verify that this is the current version before use.

MILITARY INTERESTS:

Custodians

Army - ER
Navy - YD
Air Force - 85

Review Activity

Air Force - 99

PREPARING ACTIVITY:

Navy - YD

Project 5975-0630

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:26Z
Check the source to verify that this is the current version before use.